Fiscal Policy in General Equilibrium: Model Specification and Worked Cases

A companion to the Baxter & King (1993) replication app

Abstract

This document is written for undergraduates using the interactive app at `https://baxter-king.streamlit.app/` to explore Baxter & King (1993, *American Economic Review*), "Fiscal Policy in General Equilibrium." Part I lays out the full model the app solves: preferences, technology (including the Section VI extension for productive public capital), the government's budget, the household's optimality conditions, the closed-form steady state, and the log-linear transition dynamics. Part II works through several cases that are easy to get wrong intuitively — why a spending *composition* change is not economically neutral even when public capital is unproductive, why the *financing rule* moves the economy even when every dollar of tax revenue is handed straight back, and why several steady-state variables always move by the exact same percentage by mathematical necessity, not by coincidence.

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Part I
Model specification

1 Notation

Symbol	Meaning
C_t	Consumption
N_t	Labor supplied (fraction of the time endowment); $L_t = 1 - N_t$ is leisure
Y_t	Output
K_t	Private capital stock
K_t^G	Public (government) capital stock
$I_t,\ I_t^G$	Private and public investment
$G_{T,t}$	Lump-sum government transfers
$w_t,\ r_t$	Real wage and real rental rate on capital
τ_t	Tax rate
$\theta_N,\ \theta_K = 1 - \theta_N$	Labor's and capital's shares of income
θ_G	Productivity of public capital (Section VI extension)
θ_L	Weight on leisure in utility (calibrated, not a free parameter)
δ	Depreciation rate (same for K and K^G)
A	Total factor productivity, normalized to 1

A hat over a variable, e.g. $\hat{y}_t$, denotes a percent deviation from steady state — the units used in every transition-path chart in the app.

2 Preferences

$$u(C_t, N_t) = \ln C_t + \theta_L \ln(1 - N_t) \tag{1}$$

Households value consumption and leisure, with log utility in each — the standard King–Plosser–Rebelo specification that keeps balanced-growth labor supply well behaved. Notice that government spending does *not* appear here at all: households get no direct utility from transfers $G_{T,t}$ or public investment I_t^G. This single fact is the seed of almost everything counterintuitive in Part II — government spending only ever affects the household *indirectly*, through the resource constraint and the budget constraint, never through a warm-glow "I like government spending" term.

> **Why is there no "basic government purchases" category?** Many textbook treatments split G into wasteful "purchases" G_B, productive investment I^G, and transfers G_T. Here G_B is dropped because it would be economically identical to I^G at $\theta_G = 0$: neither enters utility, and both use up real resources one-for-one. Keeping a separate G_B category would just be relabeling. Total government spending in this model is therefore always exactly two pieces: $G = I^G + G_T$.

3 Technology

$$Y_t = A\, K_t^{\theta_K}\, (K_t^G)^{\theta_G}\, N_t^{\theta_N}, \qquad \theta_K = 1 - \theta_N \tag{2}$$

Output is Cobb–Douglas in private capital and labor, exactly as in the paper's baseline model. The Section VI extension (and the app's θ_G slider) adds public capital K_t^G as a third input: it raises the productivity of every private input, without the government paying for it again next period — a non-rival "public good" flavor. Setting $\theta_G = 0$ collapses this exactly to the paper's baseline model, where public investment is a pure resource cost with *no* productivity effect. Note that $\theta_K + \theta_N = 1$ (constant returns to scale in (K, N) alone) but θ_G is *extra*: when $\theta_G > 0$ the production function has *increasing* returns to scale in (K, K^G, N) jointly, which is exactly the source of the model's "multiplier > 1" results in Part II.

4 Capital accumulation

$$K_{t+1} = (1 - \delta)K_t + I_t \qquad\qquad K_{t+1}^G = (1 - \delta)K_t^G + I_t^G \tag{3}$$

Both stocks accumulate by the standard perpetual-inventory rule and (for simplicity) depreciate at the same rate δ.

5 Resource constraint and the government budget

$$Y_t = C_t + I_t + I_t^G \tag{4}$$

Output is used for private consumption, private investment, and public investment. **Transfers do not appear here.** A transfer just moves resources from the government's books to a household's pocket without using up any output, so it nets out of the economy-wide resource constraint even though it is part of the government's budget below.

$$\tau_t Y_t = I_t^G + G_{T,t}, \qquad \tau_t = \frac{I_t^G + G_{T,t}}{Y_t} \tag{5}$$

The government's budget balances every period (no debt in this model), and the tax rate is *defined* as total government spending divided by output. This holds identically under *both* financing rules below — what differs between them is not how much revenue is raised, but whether raising it distorts the household's decisions.

$$\text{Income tax:}\quad C_t + I_t = (1-\tau_t)\big(w_t N_t + r_t K_t\big) + G_{T,t} \tag{6}$$

$$\text{Lump-sum:}\quad C_t + I_t = w_t N_t + r_t K_t - \tau_t Y_t + G_{T,t} \tag{7}$$

Under **income tax** financing, $(1-\tau_t)$ is a proportional wedge that shrinks the *marginal* return to working and saving — this is what makes the tax distortionary. Under **lump-sum** financing, the same total revenue $\tau_t Y_t$ is instead collected as a poll tax that does not depend on how much the household works or saves, so the household faces w_t and r_t directly (no $(1-\tau_t)$ term at all): a pure income (wealth) effect with no substitution effect at the margin.

6 Household optimization

Maximizing lifetime utility subject to the budget constraint gives two first-order conditions, one per financing rule.

$$\text{Labor-leisure, income tax:}\qquad \theta_L \frac{C_t}{1-N_t} = (1-\tau_t)\, w_t \tag{8}$$

$$\text{Labor-leisure, lump-sum:}\qquad \theta_L \frac{C_t}{1-N_t} = w_t \tag{9}$$

The household works until the marginal disutility of an extra hour (left side) equals its marginal after-tax benefit (right side). Under income tax financing that benefit is shrunk by $(1-\tau_t)$; under lump-sum financing there is no such wedge at all.

$$\text{Euler, income tax:}\qquad \frac{1}{C_t} = \beta\,\frac{1+(1-\tau_{t+1})\,\mathrm{MPK}_{t+1}-\delta}{C_{t+1}} \tag{10}$$

$$\text{Euler, lump-sum:}\qquad \frac{1}{C_t} = \beta\,\frac{1+\mathrm{MPK}_{t+1}-\delta}{C_{t+1}},\qquad \beta = \frac{1}{1+r} \tag{11}$$

There is no expectation operator here: this model has no stochastic element (no random shocks to technology, policy, or anything else), so every "future" variable is simply its known, perfect-foresight value. The household is not *guessing* what C_{t+1} or MPK_{t+1} will be — under a given policy path it *knows*, and the Euler equation is an ordinary deterministic difference equation linking today's consumption to tomorrow's. (This is what makes the transition path in Section 7 below solvable by exact saddle-path methods rather than by simulation.)

Prices are marginal products: $w_t = \theta_N Y_t/N_t$ and $\mathrm{MPK}_t = \theta_K Y_t/K_t$. In steady state, $(1-\tau)\mathrm{MPK} = r+\delta$ under income tax, and $\mathrm{MPK} = r+\delta$ under lump-sum.

7 Steady state (closed form)

Because $\theta_K + \theta_N = 1$, dividing the technology equation through by N leaves the capital/labor ratio $\kappa = K/N$ pinned down by a capital-Euler condition written in *per-worker* public capital K^G/N — but with an extra N^{θ_G} factor that does *not* cancel, since K^G is itself funded as a share of *aggregate* output, not of output per worker:

$$\text{Income tax:} \qquad (1-\tau)\,\theta_K\, A\,(K^G/N)^{\theta_G}\, N^{\theta_G}\, \kappa^{\theta_K - 1} = r + \delta \tag{12}$$

$$\text{Lump-sum:} \qquad \theta_K\, A\,(K^G/N)^{\theta_G}\, N^{\theta_G}\, \kappa^{\theta_K - 1} = r + \delta \tag{13}$$

When $\theta_G = 0$ the N^{θ_G} factor is just 1 and this collapses to the familiar CRS-in-(K, N) formula, solved for κ directly. When $\theta_G > 0$, $K^G = I^G/\delta$ itself depends on output, *and* the equation above depends on N directly too, so the app solves a small nested fixed point (first on κ given a guess for N, then updating N, iterating to convergence).

Everything else then follows in closed form, with $s_C = 1 - s_I - s_{IG}$ the steady-state consumption share (transfers never reduce s_C, since they are not resource-using) and $s_I = \delta\kappa/(Y/N)$ the investment share:

$$\text{Income tax:} \qquad N = \frac{(1-\tau)\,\theta_N}{\theta_L\, s_C + (1-\tau)\,\theta_N} \tag{14}$$

$$\text{Lump-sum:} \qquad N = \frac{\theta_N}{\theta_L\, s_C + \theta_N} \tag{15}$$

$$Y = \frac{Y}{N}\cdot N, \qquad K = \kappa N, \qquad C = s_C\, Y, \qquad I = s_I\, Y \tag{16}$$

θ_L itself is not a free parameter: it is *calibrated* once, at a chosen baseline policy, so that steady-state N matches a target hours-worked share (the paper's Table 1 strategy, and $N_{\text{target}} = 20\%$ in the app). Holding that calibrated θ_L fixed while comparing two different policies is what lets N move *honestly* in response to a policy change, rather than being pinned back to the target by construction.

8 Transition dynamics (log-linearized)

Around a steady state, every equation above can be approximated to first order in percent deviations. The production and labor-supply conditions combine into one static relationship,

$$\hat{y}_t = y_k\, \hat{k}_t + y_{kg}\, \hat{k}^G_t + y_c\, \hat{c}_t + y_g\, \hat{g}_t, \tag{17}$$

capital accumulation becomes a linear law of motion for the *predetermined* state $\hat{k}_t$,

$$\hat{k}_{t+1} = \Phi_{kk}\, \hat{k}_t + \Phi_{kc}\, \hat{c}_t + \Phi_{kg}\, \hat{g}_t + \Phi_{k,kg}\, \hat{k}^G_t, \tag{18}$$

and the Euler equation pins down the *forward-looking* ("jump") variable $\hat{c}_t$ in terms of tomorrow's capital and government spending,

$$\hat{c}_t = \text{coef}_c\, \hat{c}_{t+1} + \text{coef}_k\, \hat{k}_{t+1} + \text{coef}_g\, \hat{g}_{t+1} + \text{coef}_{kg}\, \hat{k}^G_{t+1}. \tag{19}$$

Public capital's own law of motion is exogenous, so the system has exactly one predetermined and one jump variable: it is **saddle-path stable**, with one stable and one unstable eigenvalue of the reduced-form transition matrix. The app diagonalizes that matrix and pins the unstable component to the present value of *future* government-spending shocks (the King–Plosser–Rebelo / Blanchard–Kahn method) rather than simulating forward naively, which would blow up numerically.

9 Reading the app: two comparison panels

Every chart in the app is one of the two objects above: the closed-form steady state of Section 7, or the log-linear transition path of this section. The app presents them as **three panels**. The first two differ only in *which two steady states are being compared* — the underlying mathematics is identical in both; the third simply reorganizes Panel 2's own comparison by variable.

9.1 Panel 1: Cumulative Effects

Compares the paper's one *fixed* benchmark calibration ($\theta_N = 0.58$, $\delta = 10\%$, $r = 6.5\%$, $G/Y = 20\%$, all of it transfers, lump-sum financing, $\theta_G = 0$) against **whatever the sidebar is currently set to**. Every control — financing, composition, θ_G, r, and G/Y — can differ between the two columns simultaneously, so this panel reports the *net, cumulative* effect of everything that has been changed at once. It is the right panel for questions like Cases A and B below: "what does switching to income tax financing do to the economy, holding everything else at the benchmark?"

Its steady-state comparison is the direct application of Section 7: solve for the benchmark's $(\kappa, N, Y, C, \ldots)$ once, solve for the current settings' $(\kappa, N, Y, C, \ldots)$ once (re-using the benchmark's calibrated θ_L on both sides, so that labor supply is free to respond honestly rather than being pinned back to the same target by construction), and report percent differences.

Its *time-series* chart applies the log-linear transition machinery of this section in a way that goes slightly beyond a simple $\Delta G/Y$ experiment: it treats the change as a **permanent, immediate switch in regime**. The economy is imagined to start ($t = 0$) with the *benchmark's* capital stocks K, K^G but the *current settings'* structural parameters and policy from that point on. Concretely, the transition is linearized around the *destination* steady state (the current settings), with the predetermined capital gap set from the origin:

$$\hat{k}_0 = \ln \frac{K_{\text{benchmark}}}{K_{\text{current}}}, \qquad \hat{k}_0^G = \ln \frac{K^G_{\text{benchmark}}}{K^G_{\text{current}}}, \tag{20}$$

and no further government-spending shock ($\hat{g}_t = 0$ for all t), since G/Y itself is assumed to jump to its new share immediately at $t = 0$ — only capital is predetermined and has to adjust gradually. This is exactly the same convention the model already uses for a *permanent* $\Delta G/Y$ shock (Panel 2 below); Panel 1 simply allows the destination regime's θ_G, financing rule, and r to differ from the origin's as well, not just G/Y.

9.2 Panel 2: The Marginal Effects of Government Spendings

Compares $G/Y = 20\%$ against **the sidebar's current** G/Y, holding financing, composition, θ_G, and r fixed at whatever the sidebar currently has them at, *on both sides* of the comparison. This isolates the effect of the $\Delta G/Y$ lever specifically: resetting G/Y to 20% always brings this panel back to exactly zero, no matter what the other controls are set to. It is the panel behind the app's headline multipliers ($\Delta Y/\Delta G$) and the Table-3-style duration-sensitivity replication.

Its time series is the ordinary $\Delta G/Y$ transition path: $\hat{g}_t$ is the (possibly temporary) government-spending shock itself, $\hat{k}_0 = 0$ (capital starts exactly at the pre-shock steady state, since only G/Y changed, nothing structural), and the linearization is around whichever steady state the shock is being compared to (the new steady state, for a permanent shock; the old one, for a temporary shock).

9.3 Panel 3: Multiplier Effects

Takes the *same* pair of steady states as Panel 2 — $G/Y = 20\%$ vs. the sidebar's current G/Y, everything else held fixed — and reports, for each of $X \in \{Y, C, I, K, K^G\}$, the multiplier

$$\frac{\Delta X}{\Delta G} = \frac{X_{\text{current}} - X_{20\%}}{G_{\text{current}} - G_{20\%}}, \tag{21}$$

both **long-run** (using the two steady-state levels directly) and **on impact** (using each variable's year-0 level along Panel 2's own transition path in place of X_{current}). No new steady state or transition path is computed here — it is purely Panel 2's comparison, reorganized by variable instead of collapsed into a single headline number. A multiplier above +1.00 for output means a marginal dollar of spending raises long-run output by *more* than a dollar, the paper's central result; investment's impact multiplier can be large even when its long-run multiplier is small, since capital only needs to move once to reach the new steady state (the "accelerator boom").

Why not just one panel? A single "current settings vs. some baseline" panel cannot serve every purpose at once. If the baseline tracks the current settings on everything except G/Y (Panel 2's convention), then changing financing *alone* — with G/Y left at 20% — shows up as *no change at all*, since both sides of the comparison would use the same financing rule. If instead the baseline is the one truly fixed benchmark (Panel 1's convention), resetting G/Y to 20% will not zero out the panel unless every other control also happens to match the benchmark. Splitting the two eliminates this ambiguity: Panel 1 answers "how far is the economy, in total, from the textbook case?"; Panel 2 answers "what, specifically, is this one lever doing, in aggregate?"; Panel 3 answers the same question as Panel 2, but "…and how does that break down variable by variable?"

Part II
Worked cases

The three cases below are the ones students most often get wrong by intuition alone. Each one follows directly from the equations in Part I; none of them requires anything beyond substituting numbers into the closed-form steady state of Section 7.

10 Case A: composition is *not* neutral once $I^G > 0$, even at $\theta_G = 0$

Claim to test: "If public capital isn't productive ($\theta_G = 0$), then it shouldn't matter whether government spending is transfers or public investment — both are just money moving around."

Why this is wrong. Transfers G_T are *not* resource-using: they cancel out of the resource constraint $Y = C + I + I^G$ entirely. Public investment I^G *is* resource-using, with or without $\theta_G > 0$ — it is a real claim on output that competes with C and I for the same Y. So shifting the composition of a fixed total G/Y from transfers toward public investment is economically identical to raising ordinary, unproductive government purchases: a negative wealth effect that makes the household work more, even though public capital does nothing for productivity.

Numerical example. $\theta_N = 0.58$, $\delta = 10\%$, $r = 6.5\%$, $G/Y = 20\%$, $\theta_G = 0$, lump-sum financing, θ_L calibrated once at $I^G/G = 0\%$ so that $N = 20.0\%$ there. This is exactly what **Panel 1 (Cumulative Effects)**, Section 9, shows if only the composition slider (item 5) is moved away from the benchmark's $I^G/G = 0\%$.

I^G/G share	N	Y	C	I^G
0% (all transfers)	20.0%	0.3934	0.2933	0.0000
50%	22.4%	0.4407	0.2845	0.0441
100% (all investment)	25.5%	0.5009	0.2732	0.1002

N, Y, and I all rise as the investment share rises, purely from the resource-cost channel — *before* any productivity effect is even switched on. This is exactly why, at $\theta_G = 0$, **Panel 2** still reports a nonzero headline multiplier $\Delta Y/\Delta G$ for an ordinary, unproductive spending increase (≈ 1.1–1.2 under lump-sum financing) — $\theta_G = 0$ turns off the *productivity* channel, not the resource-cost channel that drives this table.

The one truly neutral corner case. If (and only if) financing is lump-sum *and* the composition is already 100% transfers ($I^G/G = 0$), then changing the *level* of G/Y is a pure wash: a lump-sum tax funding an equal-sized lump-sum transfer leaves Y, C, I, K, N completely flat, since there is no $(1-\tau)$ wedge to distort anything and no resource-using spending to crowd anything out. This is the *only* combination of settings under which government spending has literally zero real effect.

11 Case B: the financing rule moves the economy even at 100% transfers

Claim to test: "If every dollar of tax revenue is handed straight back to households as a transfer, it shouldn't matter whether it was collected as a lump-sum poll tax or an income tax — the household ends up with the same net income either way."

Why this is wrong. The claim confuses the *level* of net transfers with the *marginal* incentive to work and save. Under income tax financing, the household's capital-Euler condition contains a $(1-\tau)$ wedge on the *pre-tax* marginal product of capital: $(1-\tau)\,\theta_K A\kappa^{\theta_K - 1} = r + \delta$, instead of $\theta_K A\kappa^{\theta_K - 1} = r + \delta$. To earn the same after-tax return r, the household needs a *higher* pre-tax marginal product, which means holding *less* capital per worker κ in equilibrium — a real, permanent effect on the capital stock that has nothing to do with where the tax revenue eventually lands. The same logic applies to the labor-leisure condition.

Numerical example. $\theta_N = 0.58$, $\delta = 10\%$, $r = 6.5\%$, $G/Y = 20\%$, all of it transfers ($I^G/G = 0\%$), $\theta_G = 0$. θ_L is calibrated *separately* under each financing rule so that $N = 20.0\%$ in both cases — this isolates the capital-stock channel from any mechanical change in hours. **Panel 1 (Cumulative Effects)**, Section 9, shows this same financing switch, but holds θ_L fixed at the benchmark's calibrated value throughout, so N moves too under income tax (it does not recalibrate back to 20.0%) — the mechanism is identical, only this table's presentation isolates it more cleanly.

Financing	N	Y	C	K	w
Lump-sum	20.0%	0.3934	0.2933	1.0014	1.1409
Income tax	20.0%	0.3347	0.2666	0.6816	0.9707

Switching from lump-sum to income tax financing, with G/Y, composition, and θ_G all unchanged, cuts Y by $\approx 15\%$ and K by $\approx 32\%$. N is unchanged only because θ_L was *recalibrated* to hit the same target hours under each rule — if θ_L is instead held fixed at one rule's calibrated value (as the app does when comparing two *policies* under the *same* financing rule), N falls too under income tax, since the labor-leisure wedge directly discourages work at the margin.

Takeaway. There is no "neutral financing switch" the way there is a neutral composition/level corner case in Case A. The distortion lives on the household's first-order conditions — the *margin* — not on the government's balance sheet, so it cannot be undone by any choice of what the revenue funds.

12 Case C: four steady-state variables always move by the identical percentage

Claim to test: "If two variables show the exact same percent change in a comparison table, that must be a coincidence or a bug."

Why this is wrong. Three of the model's accounting identities are exact, not approximate, and they chain together to force four variables — total government spending G, public investment I^G, transfers G_T, and public capital K^G — to move by *exactly* the same percentage whenever the composition shares and δ are held fixed between the two points being compared:

$$I^G = f_{I^G} \cdot G, \qquad G_T = f_{G_T} \cdot G \qquad \Longrightarrow \qquad \%\Delta I^G = \%\Delta G_T = \%\Delta G \tag{22}$$

$$K^G = \frac{I^G}{\delta} \qquad \Longrightarrow \qquad \%\Delta K^G = \%\Delta I^G \tag{23}$$

Chaining these: $\%\Delta K^G = \%\Delta I^G = \%\Delta G = \%\Delta G_T$, always, regardless of what is *driving* the change in G (a change in θ_G, r, financing, or G/Y itself). The *levels* of these four variables can differ enormously — K^G is typically many times larger than I^G, since $K^G = I^G/\delta$ with δ small — but the *ratio* between any two comparison points is forced to be identical by these three equations, so their percent changes print identically in any comparison table. This is a feature of the model's bookkeeping, not evidence of an error.

13 Summary table

Question	Answer
Does raising I^G/G at $\theta_G = 0$ do anything?	**Yes** — it acts like ordinary unproductive spending (Case A).
Is there *any* setting where changing G/Y does nothing at all?	**Yes, exactly one:** lump-sum financing *and* 100% transfers (Case A, notebox).
Does switching financing alone matter if all revenue is transferred back?	**Yes** — the $(1-\tau)$ wedge is on the margin, not on net income (Case B).
Why do G, I^G, G_T, K^G often show identical % change?	Because $I^G = f_{I^G}G$, $G_T = f_{G_T}G$, and $K^G = I^G/\delta$ are exact identities whenever composition shares and δ are unchanged (Case C).

Part III

Appendix: full derivations

This appendix derives, from scratch, every equation asserted without proof in Part I: the two household first-order conditions (labor-leisure and Euler) for both financing rules, and the closed-form steady-state equations built on top of them. Nothing here is new economics — it is the algebra that Part I skipped over.

A The firm's problem (why $w_t = \mathrm{MPL}_t$ and $r_t = \mathrm{MPK}_t$)

Before touching the household, it is worth pinning down where the prices w_t and r_t in the household's budget constraint actually come from. A representative, perfectly competitive firm rents private capital and labor from households each period and uses the publicly provided capital stock K_t^G for free (it is non-rival and supplied by the government, not purchased in a market). The firm solves, period by period,

$$\max_{K_t,\, N_t} \quad A\, K_t^{\theta_K} (K_t^G)^{\theta_G} N_t^{\theta_N} \; - \; w_t N_t \; - \; r_t K_t. \tag{24}$$

Since K_t^G is taken as given (external to the firm's own choice), the first-order conditions are the ordinary marginal-product conditions:

$$\frac{\partial Y_t}{\partial N_t} = w_t \implies w_t = \theta_N \, \frac{Y_t}{N_t} \equiv \mathrm{MPL}_t, \tag{25}$$

$$\frac{\partial Y_t}{\partial K_t} = r_t \implies r_t = \theta_K \, \frac{Y_t}{K_t} \equiv \mathrm{MPK}_t. \tag{26}$$

Zero profit follows automatically from constant returns to scale in (K_t, N_t) (Euler's homogeneous-function theorem): at these prices the firm earns exactly $\theta_K Y_t + \theta_N Y_t = Y_t$, paying out all of output to the two rented factors. This is why r_t (the household's capital income rate) and MPK_t can be used interchangeably below — they are the *same number* in every period, by the firm's optimality condition, not by assumption.

B Setting up the household's problem

The household chooses sequences $\{C_t, N_t, K_{t+1}\}_{t=0}^{\infty}$ to maximize discounted lifetime utility,

$$\max \sum_{t=0}^{\infty} \beta^t \Big[\ln C_t + \theta_L \ln(1 - N_t) \Big], \qquad \beta = \frac{1}{1+r}, \tag{27}$$

subject to one budget constraint per period and the capital accumulation identity $K_{t+1} = (1 - \delta)K_t + I_t$, i.e. $I_t = K_{t+1} - (1-\delta)K_t$. Substituting this into the two budget constraints of Section 5 gives, respectively,

$$\text{Income tax:} \qquad C_t + K_{t+1} - (1-\delta)K_t = (1-\tau_t)\big(w_t N_t + r_t K_t\big) + G_{T,t}, \tag{28}$$

$$\text{Lump-sum:} \qquad C_t + K_{t+1} - (1-\delta)K_t = w_t N_t + r_t K_t - \tau_t Y_t + G_{T,t}. \tag{29}$$

There is no expectation operator anywhere in this problem: the model has no stochastic element, so the household is choosing a sequence under **perfect foresight** — it knows $w_t, r_t, \tau_t, G_{T,t}$ (and hence Y_t) at every future date along the equilibrium path, given the policy being analyzed. Two modeling conventions are worth flagging explicitly:

(i) Under *income tax* financing, the wedge $(1 - \tau_t)$ multiplies the household's *own* factor income $w_t N_t + r_t K_t$ — it is a genuine proportional tax on the household's choices, so it will show up when we differentiate with respect to N_t and K_{t+1} below.

(ii) Under *lump-sum* financing, the household's tax bill is written as $\tau_t Y_t$, using *aggregate* output Y_t, not the household's own income. The (representative) household takes $\tau_t Y_t$ as a parameter it cannot affect with its own N_t or K_t choice — exactly what makes a lump-sum tax lump-sum. This is why τ_t never appears in the lump-sum first-order conditions below, even though $\tau_t Y_t > 0$ in general.

C Deriving the labor-leisure and Euler conditions

Attach a Lagrange multiplier $\lambda_t \geq 0$ (period-t's marginal utility of an extra unit of income) to the period-t budget constraint and form the Lagrangian. Under **income tax** financing,

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ \ln C_t + \theta_L \ln(1 - N_t) + \lambda_t \Big[(1-\tau_t)(w_t N_t + r_t K_t) + G_{T,t} - C_t - K_{t+1} + (1-\delta)K_t \Big] \Big\}. \tag{30}$$

Step 1: the consumption FOC. Differentiating with respect to C_t (which appears only in period t's bracket) gives

$$\frac{\partial \mathcal{L}}{\partial C_t} = \beta^t \left[\frac{1}{C_t} - \lambda_t \right] = 0 \implies \lambda_t = \frac{1}{C_t}. \tag{31}$$

This is the standard result that the multiplier on the budget constraint equals the marginal utility of consumption — an extra unit of income is worth exactly what an extra unit of consumption is worth, since the household can always convert one into the other one-for-one within a period.

Step 2: the labor-leisure FOC. Differentiating with respect to N_t (also period-t-only) gives

$$\frac{\partial \mathcal{L}}{\partial N_t} = \beta^t \left[-\frac{\theta_L}{1 - N_t} + \lambda_t (1 - \tau_t) w_t \right] = 0. \tag{32}$$

(The $-\theta_L/(1 - N_t)$ term has a sign flip built in: $\partial \ln(1 - N_t)/\partial N_t = -1/(1 - N_t)$.) Solving and substituting $\lambda_t = 1/C_t$ from Step 1,

$$\frac{\theta_L}{1 - N_t} = \frac{(1 - \tau_t) w_t}{C_t} \implies \boxed{\theta_L \, \frac{C_t}{1 - N_t} = (1 - \tau_t)\, w_t.} \tag{33}$$

This is exactly the labor-leisure condition of Section 6. Economically: the left side is the marginal rate of substitution between consumption and leisure (how many units of consumption the household would give up for one more unit of leisure); the right side is the after-tax wage, the market price of leisure in consumption units. Optimality equates the two.

Step 3: the Euler (capital) FOC. K_{t+1} is the one variable that appears in *two* budget constraints: negatively in period t's (it is being accumulated) and positively in period $t + 1$'s (it earns capital income and is then reduced by depreciation and reinvestment). Collecting both appearances,

$$\frac{\partial \mathcal{L}}{\partial K_{t+1}} = \beta^t(-\lambda_t) + \beta^{t+1} \lambda_{t+1} \Big[(1 - \tau_{t+1}) r_{t+1} + (1 - \delta) \Big] = 0. \tag{34}$$

Dividing through by β^t and rearranging,

$$\lambda_t = \beta \, \lambda_{t+1} \Big[1 + (1 - \tau_{t+1}) r_{t+1} - \delta \Big]. \tag{35}$$

Substituting $\lambda_t = 1/C_t$, $\lambda_{t+1} = 1/C_{t+1}$, and $r_{t+1} = \mathrm{MPK}_{t+1}$ (Appendix A),

$$\boxed{\frac{1}{C_t} = \beta \, \frac{1 + (1 - \tau_{t+1})\, \mathrm{MPK}_{t+1} - \delta}{C_{t+1}}.} \tag{36}$$

This is exactly the income-tax Euler equation of Section 6. The bracketed term $1 + (1 - \tau_{t+1})\mathrm{MPK}_{t+1} - \delta$ is the after-tax gross return on one unit of capital held from t to $t + 1$: it pays $(1 - \tau_{t+1})\mathrm{MPK}_{t+1}$ in after-tax rental income and loses δ to depreciation, on top of the principal. The household is indifferent at the margin between consuming a unit today (worth $1/C_t$ in utility, by Step 1) and investing it to consume the gross return tomorrow (worth $\beta \times \text{return} \times 1/C_{t+1}$ in discounted utility).

The lump-sum case. Repeat Steps 2–3 with the lump-sum Lagrangian,

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ \ln C_t + \theta_L \ln(1 - N_t) + \lambda_t \Big[w_t N_t + r_t K_t - \tau_t Y_t + G_{T,t} - C_t - K_{t+1} + (1 - \delta) K_t \Big] \Big\}. \tag{37}$$

Because $\tau_t Y_t$ does not depend on N_t or K_{t+1} (point (ii) of Appendix B), it simply drops out when differentiating:

$$\frac{\partial \mathcal{L}}{\partial N_t} = \beta^t \left[-\frac{\theta_L}{1 - N_t} + \lambda_t w_t \right] = 0 \implies \boxed{\theta_L \, \frac{C_t}{1 - N_t} = w_t,} \tag{38}$$

$$\frac{\partial \mathcal{L}}{\partial K_{t+1}} = \beta^t(-\lambda_t) + \beta^{t+1} \lambda_{t+1} \big[r_{t+1} + (1 - \delta) \big] = 0 \implies \boxed{\frac{1}{C_t} = \beta \, \frac{1 + \mathrm{MPK}_{t+1} - \delta}{C_{t+1}}.} \tag{39}$$

No $(1-\tau)$ wedge appears in either condition: the household's marginal trade-offs between consumption/leisure and consumption today/tomorrow are completely undistorted, even though the household *does* pay $\tau_t Y_t$ in taxes every period. This is precisely the sense in which a lump-sum tax is non-distorting — it changes the household's wealth, but not the *price* of working an extra hour or saving an extra dollar.

D Deriving the closed-form steady state

A steady state is a situation where every "great ratio" and every level variable is constant over time: $C_t = C$, $N_t = N$, $K_t = K$, $Y_t = Y$, and so on, for all t. This section derives every equation of Section 7 in order, starting from the FOCs of Appendix C.

D.1 Step 1: the capital-Euler condition pins down $\kappa = K/N$

In steady state $C_t = C_{t+1} = C$ and $\text{MPK}_t = \text{MPK}_{t+1} = \text{MPK}$, so the Euler equation collapses to an equation with no time subscripts at all. Income tax case:

$$1 = \beta\big[1 + (1-\tau)\text{MPK} - \delta\big] \implies (1-\tau)\text{MPK} = \frac{1}{\beta} - 1 + \delta. \tag{40}$$

Recall $\beta = 1/(1+r)$ was *chosen* so that $1/\beta = 1 + r$ (the household's steady-state discount rate is calibrated to match a target real interest rate r). Substituting,

$$\boxed{(1-\tau)\,\text{MPK} = r + \delta.} \tag{41}$$

The lump-sum case is identical without the $(1-\tau)$ factor: $\boxed{\text{MPK} = r + \delta.}$ These are the two boxed conditions quoted without proof at the start of Section 7.

Now expand $\text{MPK} = \theta_K Y/K$ using the technology equation, and substitute $K = \kappa N$ (definition of κ):

$$\text{MPK} = \theta_K\,\frac{AK^{\theta_K}(K^G)^{\theta_G}N^{\theta_N}}{K} = \theta_K A\,K^{\theta_K-1}(K^G)^{\theta_G}N^{\theta_N} = \theta_K A\,\kappa^{\theta_K-1}N^{\theta_K-1}(K^G)^{\theta_G}N^{\theta_N}. \tag{42}$$

Since $\theta_K + \theta_N = 1$ (constant returns to scale in (K, N) alone, Section 4), the two powers of N combine to $N^{\theta_K-1+\theta_N} = N^0 = 1$, leaving

$$\text{MPK} = \theta_K A\,\kappa^{\theta_K-1}\,(K^G)^{\theta_G}. \tag{43}$$

Notice this uses the *raw* level K^G, not K^G/N: no N survives the cancellation above. To match the per-worker form used in Section 7 and in the app, multiply and divide by N^{θ_G}:

$$(K^G)^{\theta_G} = \left(\frac{K^G}{N}\right)^{\theta_G} N^{\theta_G} \implies \text{MPK} = \theta_K A\left(\frac{K^G}{N}\right)^{\theta_G} N^{\theta_G}\,\kappa^{\theta_K-1}. \tag{44}$$

Substituting into the boxed conditions above reproduces exactly

$$\text{Income tax:}\quad (1-\tau)\,\theta_K\,A\,(K^G/N)^{\theta_G}\,N^{\theta_G}\,\kappa^{\theta_K-1} = r + \delta, \tag{45}$$

$$\text{Lump-sum:}\quad \theta_K\,A\,(K^G/N)^{\theta_G}\,N^{\theta_G}\,\kappa^{\theta_K-1} = r + \delta. \tag{46}$$

Why bother with the $(K^G/N)^{\theta_G} N^{\theta_G}$ split instead of just writing $(K^G)^{\theta_G}$? Because in the fixed-point iteration the app actually solves numerically, K^G is not known in advance when $\theta_G > 0$ — it depends on output through $K^G = I^G/\delta$, which depends on κ, which is what we are solving for. Working in per-worker terms (K^G/N) lets the iteration guess and update a *scale-free ratio* first, and then multiply back in the explicit N^{θ_G} correction, rather than guessing the level K^G directly. When $\theta_G = 0$, $N^{\theta_G} = 1$ and this distinction is invisible; it only matters once public capital is productive.

When $\theta_G = 0$, equation (43) has no K^G dependence at all, and can be solved for κ directly:

$$\theta_K A\, \kappa^{\theta_K - 1} = \frac{r+\delta}{\text{tax\_factor}} \implies \kappa = \left(\frac{\text{tax\_factor} \cdot \theta_K A}{r+\delta}\right)^{\frac{1}{1-\theta_K}}, \tag{47}$$

where tax_factor $= (1-\tau)$ under income tax and 1 under lump-sum. When $\theta_G > 0$, the same formula holds but with A replaced by $A\,(K^G/N)^{\theta_G} N^{\theta_G}$, which itself depends on κ through $K^G = I^G/\delta$ and $Y = \alpha N$ — hence the nested fixed-point iteration described in Section 7.

D.2 Step 2: the investment and consumption shares

With κ (and hence $Y/N \equiv \alpha$) in hand, define the steady-state investment share directly from the capital accumulation identity $K_{t+1} = (1-\delta)K_t + I_t$: in steady state $K_{t+1} = K_t = K$, so $I = \delta K$. Dividing by Y,

$$s_I \equiv \frac{I}{Y} = \frac{\delta K}{Y} = \frac{\delta \kappa N}{\alpha N} = \frac{\delta\kappa}{\alpha}. \tag{48}$$

The resource constraint $Y = C + I + I^G$ (Section 5), divided by Y, gives $1 = s_C + s_I + s_{I^G}$, i.e.

$$\boxed{s_C = 1 - s_I - s_{I^G}.} \tag{49}$$

Transfers G_T never appear in this identity, because they were never in the resource constraint to begin with (Section 5) — this is the formal statement of "transfers are not resource-using."

D.3 Step 3: the labor-leisure condition pins down N

Start from the steady-state labor-leisure condition (income tax case),

$$\theta_L \frac{C}{1-N} = (1-\tau)\, w, \qquad w = \theta_N \frac{Y}{N} \;\; \text{(Appendix A)}. \tag{50}$$

Multiply both sides by $(1-N)$ and divide both sides by Y, using $C/Y = s_C$:

$$\theta_L\, s_C = (1-\tau)\,\theta_N\, \frac{1-N}{N}. \tag{51}$$

Multiply both sides by N:

$$\theta_L\, s_C\, N = (1-\tau)\,\theta_N\,(1-N) = (1-\tau)\theta_N - (1-\tau)\theta_N\, N. \tag{52}$$

Collect every term containing N on the left:

$$N\big[\theta_L\, s_C + (1-\tau)\theta_N\big] = (1-\tau)\theta_N. \tag{53}$$

Divide through:

$$\boxed{N = \frac{(1-\tau)\,\theta_N}{\theta_L\, s_C + (1-\tau)\,\theta_N}.} \tag{54}$$

The lump-sum case follows identically with tax_factor $= 1$ in place of $(1-\tau)$ throughout (there is no wedge in the lump-sum labor-leisure condition to begin with):

$$\boxed{N = \frac{\theta_N}{\theta_L\, s_C + \theta_N}.} \tag{55}$$

These reproduce exactly the two boxed N-equations of Section 7. Notice the entire derivation only used *ratios* ($s_C = C/Y$, θ_L, θ_N, τ) — this is why N can be solved before Y, C, or K are known in levels: once κ fixes $\alpha = Y/N$ and s_C (Steps 1–2), N falls out of a single algebraic equation, and every level variable is then a trivial multiple of N:

$$Y = \alpha N, \qquad K = \kappa N, \qquad C = s_C\, Y, \qquad I = s_I\, Y, \qquad I^G = s_{I^G}\, Y, \qquad G_T = s_{G_T}\, Y. \tag{56}$$

Where does θ_L come from? θ_L is not observed directly; it is *calibrated* by inverting the boxed N-equation at a chosen baseline policy and a target hours-worked share N_{target} (Baxter & King's Table 1 strategy). Solving the income-tax N-equation for θ_L instead of N:

$$\theta_L = \frac{(1-\tau)\,\theta_N\,(1-N_{\text{target}})}{s_C\, N_{\text{target}}}, \tag{57}$$

and analogously without the $(1-\tau)$ factor under lump-sum. This is exactly `calibrate_theta_L` in the code: it is the *same* equation as Step 3 above, just solved for the other unknown.